EXAMPLE 4 If each cell of a tumor has two targets, then the two target one hit surviving fraction is given by

$$f(x) = 1 - (1 - e^{-kx})^2,$$

where k is the average size of a cell. Analyze the graph of f to determine what effect increasing the dosage x has on decreasing the surviving fraction of tumor cells.

SOLUTION First note that if $x = 0$, then $f(0) = 1$; that is, if there is no dose, then all cells survive. Differentiating, we obtain

$$\begin{aligned} f'(x) &= 0 - 2(1 - e^{-kx})\, D_x(1 - e^{-kx}) \\ &= -2(1 - e^{-kx})(ke^{-kx}) \\ &= -2ke^{-kx}(1 - e^{-kx}). \end{aligned}$$

Since $f'(x) < 0$ for every $x > 0$ and $f'(0) = 0$, the function f is decreasing and the graph has a horizontal tangent line at the point (0, 1). We may verify that the second derivative is

$$f''(x) = 2k^2e^{-kx}(1 - 2e^{-kx}).$$

We see that $f''(x) = 0$ if $1 - 2e^{-kx} = 0$ (that is, if $e^{-kx} = \frac{1}{2}$, or, equivalently, $-kx = \ln \frac{1}{2} = -\ln 2$). This gives us

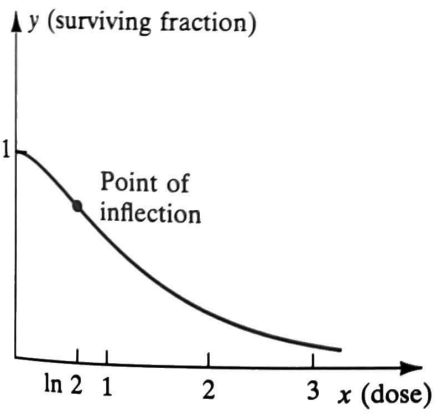
$$x = \frac{1}{k} \ln 2.$$

It can be verified that if $0 \le x < (1/k) \ln 2$, then $f''(x) < 0$, and hence the graph is concave downward. If $x > (1/k) \ln 2$, then $f''(x) > 0$, and the graph is concave upward. This implies that there is a point of inflection with x-coordinate $(1/k) \ln 2$. The y-coordinate of this point is

$$\begin{aligned} f\left(\frac{1}{k} \ln 2\right) &= 1 - (1 - e^{-\ln 2})^2 \\ &= 1 - (1 - \tfrac{1}{2})^2 = \tfrac{3}{4}. \end{aligned}$$

The graph is sketched in Figure 7.14 for the case $k = 1$. The *shoulder* on the curve near the point (0, 1) represents the threshold nature of the treatment; that is, a small dose results in very little tumor elimination. Note that if x is large, then an increase in dosage has little effect on the surviving fraction. To determine the ideal dose that should be administered to a given patient, specialists in radiation therapy must also take into account the number of healthy cells that are killed during a treatment.

FIGURE 7.14
Surviving fraction of tumor cells after a radiation treatment



EXERCISES 7.3

Exer. 1–30: Find $f'(x)$ if $f(x)$ equals the given expression.

1 e^{-5x}

2 e^{3x}

3 e^{3x^2}

4 e^{1-x^3}

5 $\sqrt{1 + e^{2x}}$

6 $1/(e^x + 1)$

7 $e^{\sqrt{x+1}}$

8 xe^{-x}

9 x^2e^{-2x}

10 $\sqrt{e^{2x} + 2x}$

11 $e^x/(x^2 + 1)$

12 $x/e^{(x^2)}$

13 $(e^{4x} - 5)^3$

14 $(e^{3x} - e^{-3x})^4$

15 $e^{1/x} + (1/e^x)$

16 $e^{\sqrt{x}} + \sqrt{e^x}$

17 $\dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

18 $e^{x \ln x}$

19 $e^{-2x} \ln x$

20 $\ln e^x$

21 $\sin e^{5x}$

22 $e^{\sin 5x}$

23 $\ln \cos e^{-x}$

24 $e^{-3x} \cos 3x$

25 $e^{3x} \tan \sqrt{x}$

26 $\sec e^{-2x}$

27 $\sec^2 (e^{-4x})$

28 $e^{-x} \tan^2 x$

29 $xe^{\cot x}$

30 $\ln (\csc e^{3x})$

Exer. 31–34: Use implicit differentiation to find y'.

31 $e^{xy} - x^3 + 3y^2 = 11$

32 $xe^y + 2x - \ln (y + 1) = 3$

33 $e^x \cot y = xe^{2y}$

34 $e^x \cos y = xe^y$

35 Find an equation of the tangent line to the graph of $y = (x - 1)e^x + 3 \ln x + 2$ at the point $P(1, 2)$.

36 Find an equation of the tangent line to the graph of $y = x - e^{-x}$ that is parallel to the line $6x - 2y = 7$.

Exer. 37–42: Find the local extrema of f. Determine where f is increasing or is decreasing, discuss concavity, find the points of inflection, and sketch the graph of f.

37 $f(x) = xe^x$

38 $f(x) = x^2e^{-2x}$

39 $f(x) = e^{1/x}$

40 $f(x) = xe^{-x}$

41 $f(x) = x \ln x$

42 $f(x) = (1 - \ln x)^2$

43 A radioactive substance decays according to the formula $q(t) = q_0e^{-ct}$, where q_0 is the initial amount of the substance, c is a positive constant, and $q(t)$ is the amount remaining after time t. Show that the rate at which the substance decays is proportional to $q(t)$.

44 The current $I(t)$ at time t in an electrical circuit is given by $I(t) = I_0e^{-Rt/L}$, where R is the resistance, L is the inductance, and I_0 is the current at time $t = 0$. Show that the rate of change of the current at any time t is proportional to $I(t)$.

45 If a drug is injected into the bloodstream, then its concentration t minutes later is given by

$$C(t) = \frac{k}{a - b}(e^{-bt} - e^{-at})$$

for positive constants a, b, and k.

(a) At what time does the maximum concentration occur?

(b) What can be said about the concentration after a long period of time?

46 If a beam of light that has intensity k is projected vertically downward into water, then its intensity $I(x)$ at a depth of x meters is $I(x) = ke^{-1.4x}$.

(a) At what rate is the intensity changing with respect to depth at 1 meter? 5 meters? 10 meters?

(b) At what depth is the intensity one-half its value at the surface? One-tenth its value?

47 The *Jenss model* is generally regarded as the most accurate formula for predicting the height of a preschooler. If $h(x)$ denotes the height (in centimeters) at age x (in years) for $\frac{1}{4} \le x \le 6$, then $h(x)$ can be approximated by

$$h(x) = 79.041 + 6.39x - e^{3.261 - 0.993x}.$$

(Compare with Example 7 of Section 7.2.)

(a) Predict the height and rate of growth when a child reaches the age of 1.

(b) When is the rate of growth largest, and when is it smallest?

48 For a population of female African elephants, the weight $W(t)$ (in kilograms) at age t (in years) may be approximated by a *von Bertanlanffy growth function* W such that

$$W(t) = 2600(1 - 0.51e^{-0.075t})^3.$$

(a) Approximate the weight and the rate of growth of a newborn.

(b) Assuming that an adult female weighs 1800 kilograms, estimate her age and her rate of growth at present.

(c) Find and interpret $\lim_{t \to \infty} W(t)$.

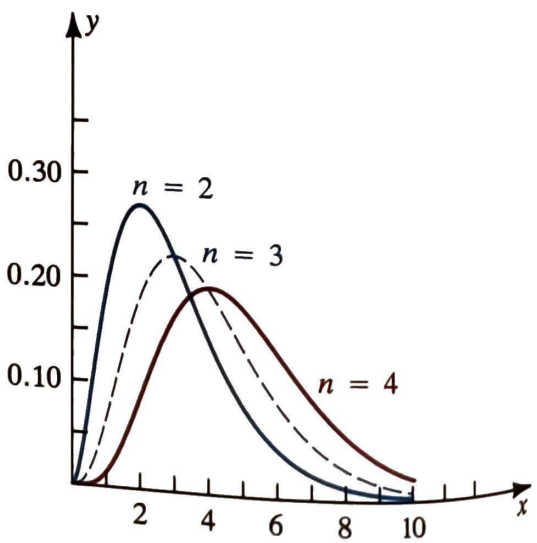
(d) Show that the rate of growth is largest between the ages of 5 years and 6 years.

49 Gamma distributions, which are important in traffic control studies and probability theory, are determined by $f(x) = cx^ne^{-ax}$ for $x > 0$, a positive integer n, a positive constant a, and $c = a^{n+1}/n!$. Shown in the figure are graphs corresponding to $a = 1$ for $n = 2, 3$, and 4.

(a) Show that f has exactly one local maximum.

(b) If $n = 4$, determine where $f(x)$ is increasing most rapidly.

EXERCISE 49



50 The relative number of gas molecules in a container that travel at a velocity of v cm/sec can be computed by means of the *Maxwell-Boltzmann speed distribution*

$F(v) = cv^2 e^{-mv^2/(2kT)}$, where T is the temperature (in °K), m is the mass of a molecule, and c and k are positive constants. Show that the maximum value of F occurs when $v = \sqrt{2kT/m}$.

51 An *urban density model* is a formula that relates the population density (in number/mi²) to the distance r (in miles) from the center of the city. The formula $D = ae^{-br+cr^2}$, where a, b, and c are positive constants, has been found to be appropriate for certain cities. Determine the shape of the graph for $r \geq 0$.

52 The effect of light on the rate of photosynthesis can be described by

$$f(x) = x^a e^{(a/b)(1-x^b)}$$

for $x > 0$ and positive constants a and b.

(a) Show that f has a maximum at $x = 1$.

(b) Conclude that if $x_0 > 0$ and $y_0 > 0$, then $g(x) = y_0 f(x/x_0)$ has a maximum $g(x_0) = y_0$.

53 The rate R at which a tumor grows is related to its size x by the equation $R = rx \ln (K/x)$, where r and K are positive constants. Show that the tumor is growing most rapidly when $x = e^{-1}K$.

54 If p denotes the selling price (in dollars) of a commodity and x is the corresponding demand (in number sold per day), then the relationship between p and x may be given by $p = p_0 e^{-ax}$ for positive constants p_0 and a. Suppose $p = 300e^{-0.02x}$. Find the selling price that will maximize daily revenues (see page 228).

55 In statistics the probability density function for the normal distribution is defined by

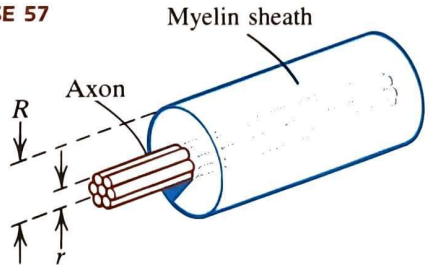
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2} \quad \text{with} \quad z = \frac{x-\mu}{\sigma}$$

for real numbers μ and $\sigma > 0$ (μ is the *mean* and σ^2 is the *variance* of the distribution). Find the local extrema of f, and determine where f is increasing or is decreasing. Discuss concavity, find points of inflection, find $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$, and sketch the graph of f (see Example 3).

C 56 The integral $\int_a^b e^{-x^2}\,dx$ has applications in statistics. Use the trapezoidal rule, with $n = 10$, to approximate this integral if $a = 0$ and $b = 1$.

57 Nerve impulses in the human body travel along nerve fibers that consist of an *axon*, which transports the impulse, and an insulating coating surrounding the axon, called the *myelin sheath* (see figure). The nerve fiber is similar to an insulated cylindrical cable, for which the velocity v of an impulse is given by $v = -k(r/R)^2 \ln (r/R)$, where r is the radius of the cable and R is the insulation radius. Find the value of r/R that maximizes v. (In most nerve fibers $r/R \approx 0.6$.)

EXERCISE 57



58 Sketch the graph of the *three target–one hit surviving fraction* (with $k = 1$) given by $f(x) = 1 - (1 - e^{-x})^3$ (see Example 4).

C **Exer. 59–60: Use Newton's method to approximate the real root of the equation to two decimal places.**

59 $e^{-x} = x$ 60 $x \ln x = 1$

7.4 INTEGRATION

We may use differentiation formulas for ln to obtain formulas for integration. In particular, by Theorem (7.11),

$$D_x \ln |g(x)| = \frac{1}{g(x)} g'(x),$$

which gives us the integration formula

$$\int \frac{1}{g(x)} g'(x)\,dx = \ln |g(x)| + C.$$

This is restated in the next theorem in terms of the variable u.